

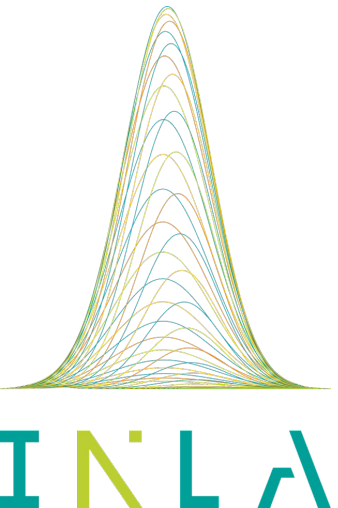


A new Bayesian approach to modelling of short and long term earthquake forecasting using inlabru

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Researchers: Kirsty Bayliss, Francesco Serafini, Farnaz Kamranzad

Faculty: Finn Lindgren, Ian Main



Engineering and
Physical Sciences
Research Council



Inla + inlabru

<https://www.r-inla.org>:

- The integrated nested Laplace approximation (INLA) is a method for approximate Bayesian inference.
 - An alternative to other methods such as MCMC because of its speed and ease of use via the R-INLA package

<https://inlabru-org.github.io/inlabru/>:

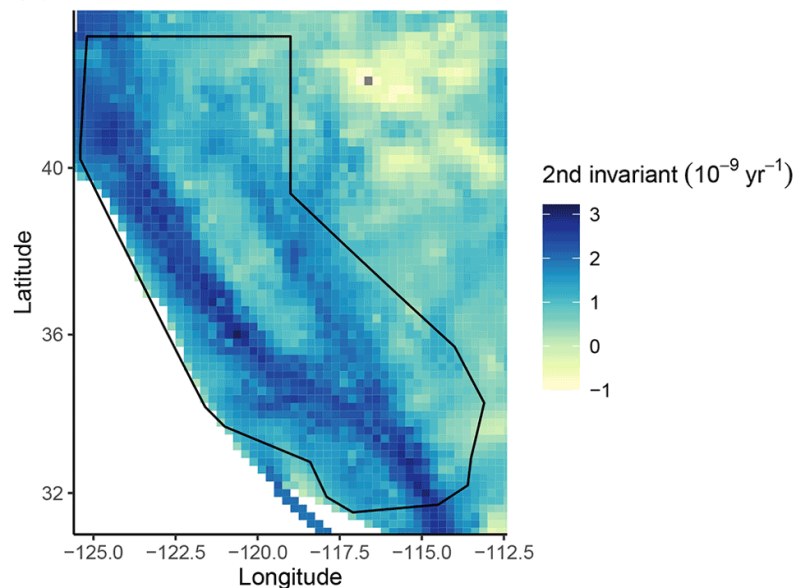
- The goal of [inlabru](https://inlabru-org.github.io/inlabru/) is to facilitate spatial modeling using integrated nested Laplace approximation via the [R-INLA package](https://www.r-inla.org).

1. Time independent spatial models: California (Log-Gaussian Cox process)

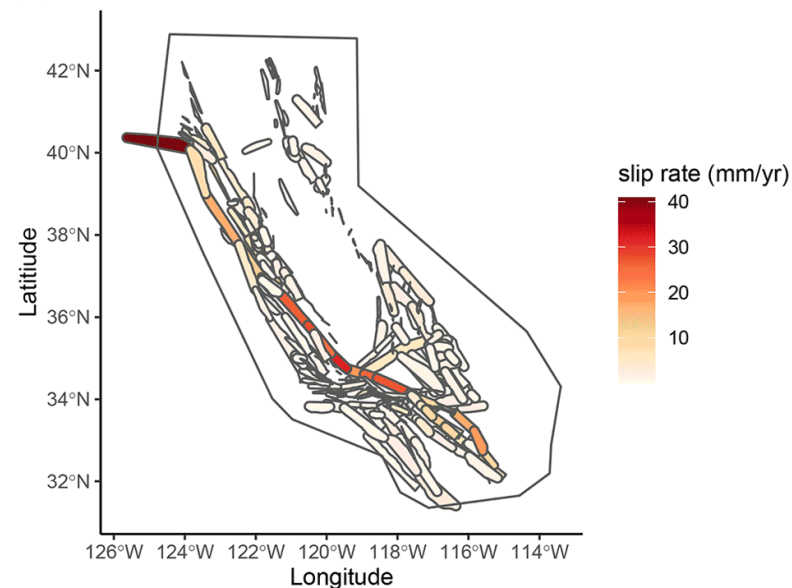


1. Bayliss, Naylor, Illian & Main (2020). Data-driven optimization of seismicity models using diverse data sets: Generation, evaluation, and ranking using inlabru. *Journal of Geophysical Research: Solid Earth*, 125, e2020JB020226. <https://doi.org/10.1029/2020JB020226>
2. Bayliss, Naylor, Kamranzad, and Main (2022) Pseudo-prospective testing of 5-year earthquake forecasts for California using inlabru, *Nat. Hazards Earth Syst. Sci.*, 22, 3231–3246, 2022 <https://doi.org/10.5194/nhess-22-3231-2022>

(a) log strain rate



(b) Slip rates (NeoKinema model)



Intensity is function of a linear predictor

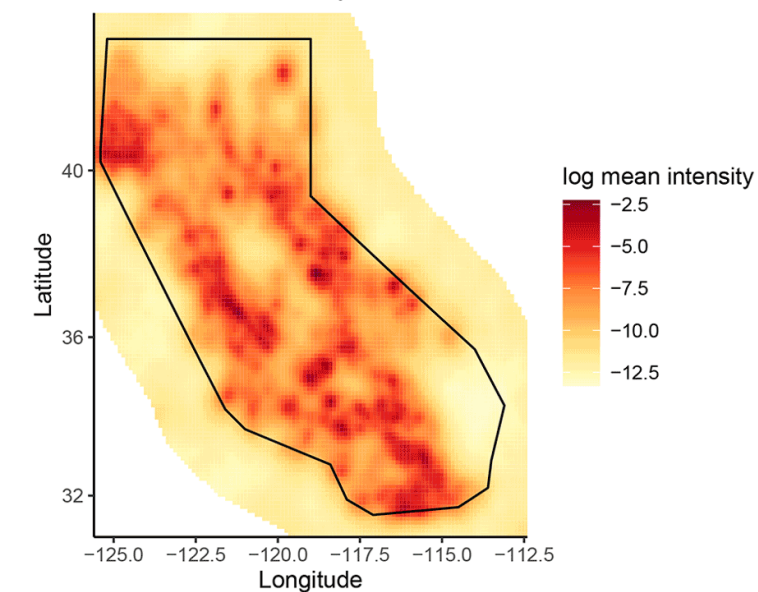
$$\lambda(s) = \exp(-\eta(s))$$

Linear predictor combines:

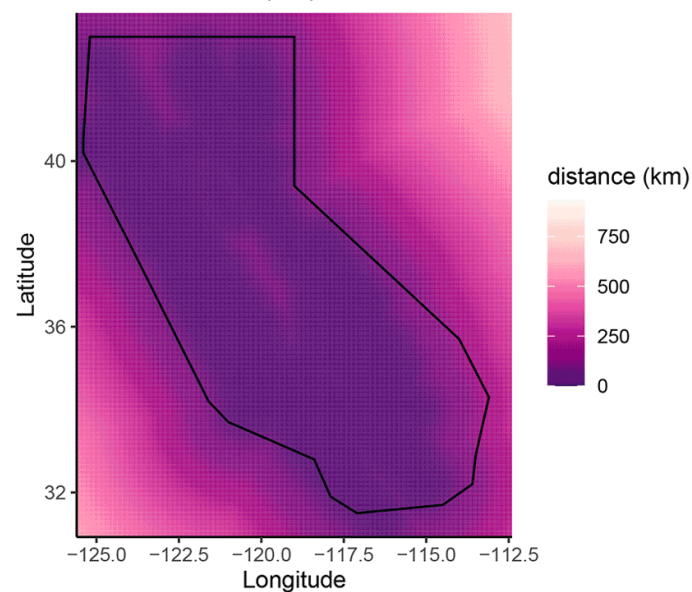
- Intercept
- Weighted sum of spatial covariates
- Gaussian Markov Random Field

$$\eta(s) = \beta_0 + \sum_{m=1}^M \beta_m x_m(s) + \zeta(s)$$

(c) Smoothed seismicity



(d) Fault Distance (km)



Gaussian Markov Random Field describes correlation in the spatial residuals

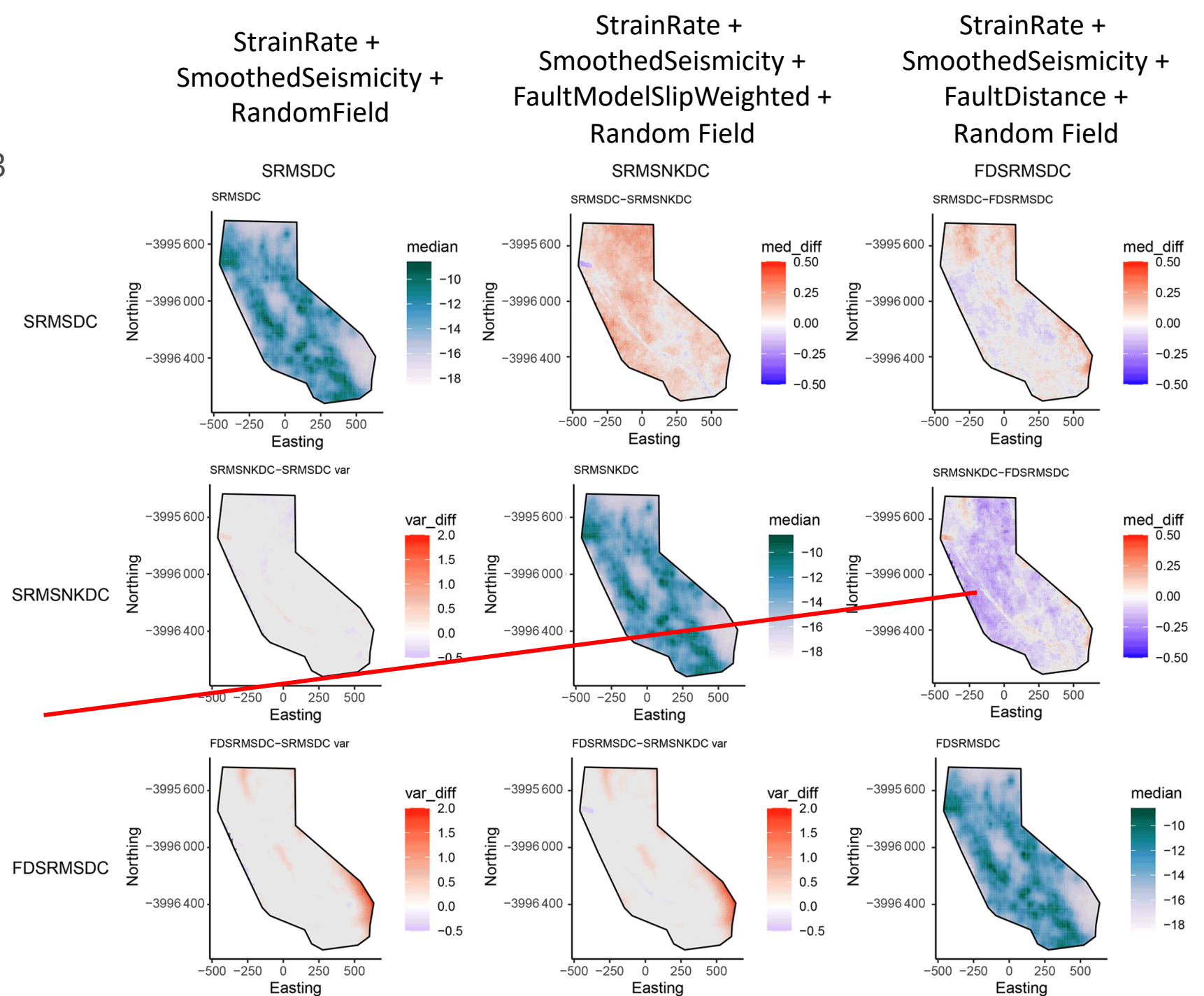
- Unmodelled clustering
- Location uncertainty
- Unseen trigger

Catalogue input from the UCERF3

- California earthquakes
- 20 years (1985–2005)
- magnitude ≥ 4.95
- Declassified

Forecasts of rates for $M > 4.95$

Median difference shows
influence of fault network



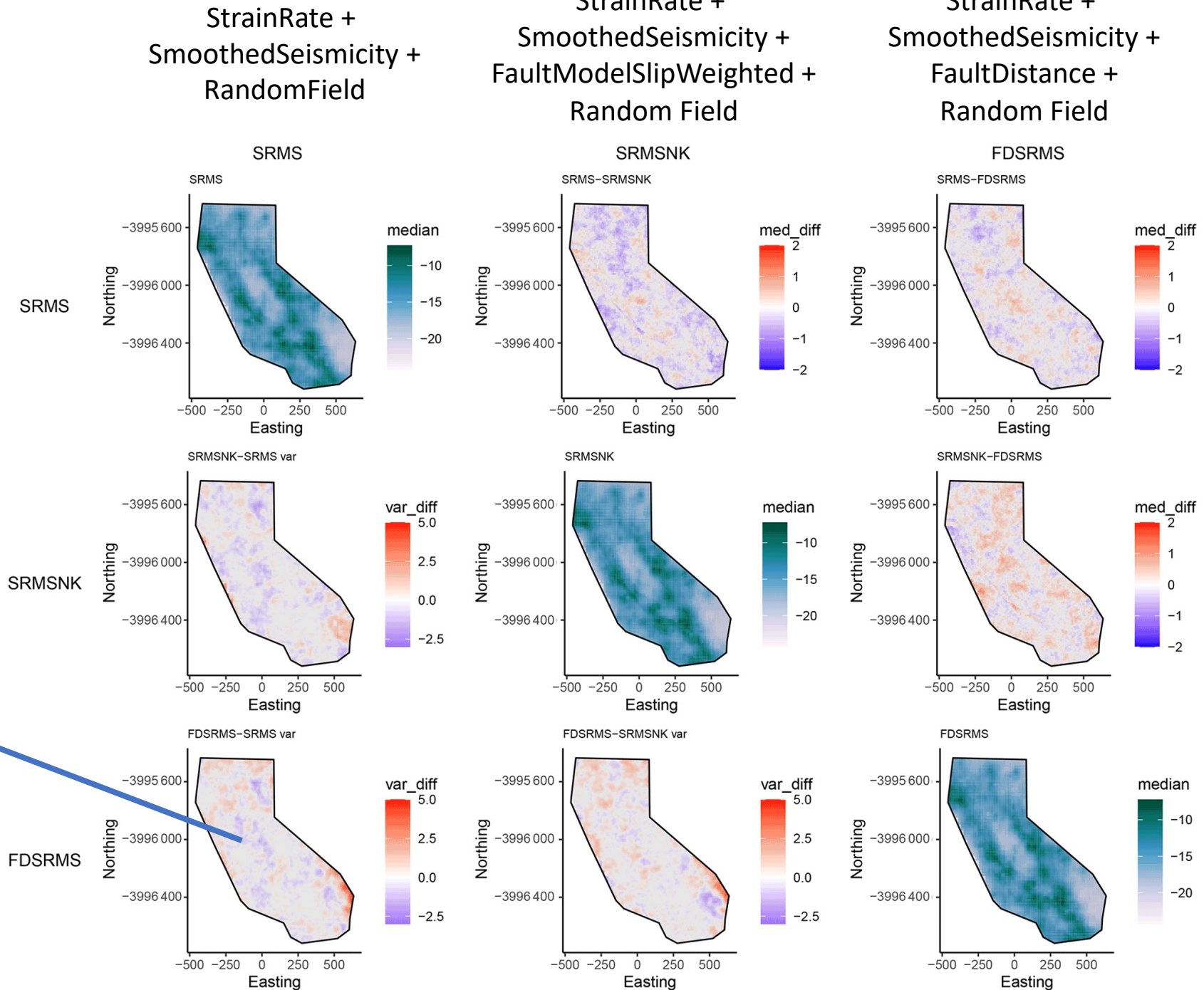
Catalogue input from the UCERF3

- California earthquakes
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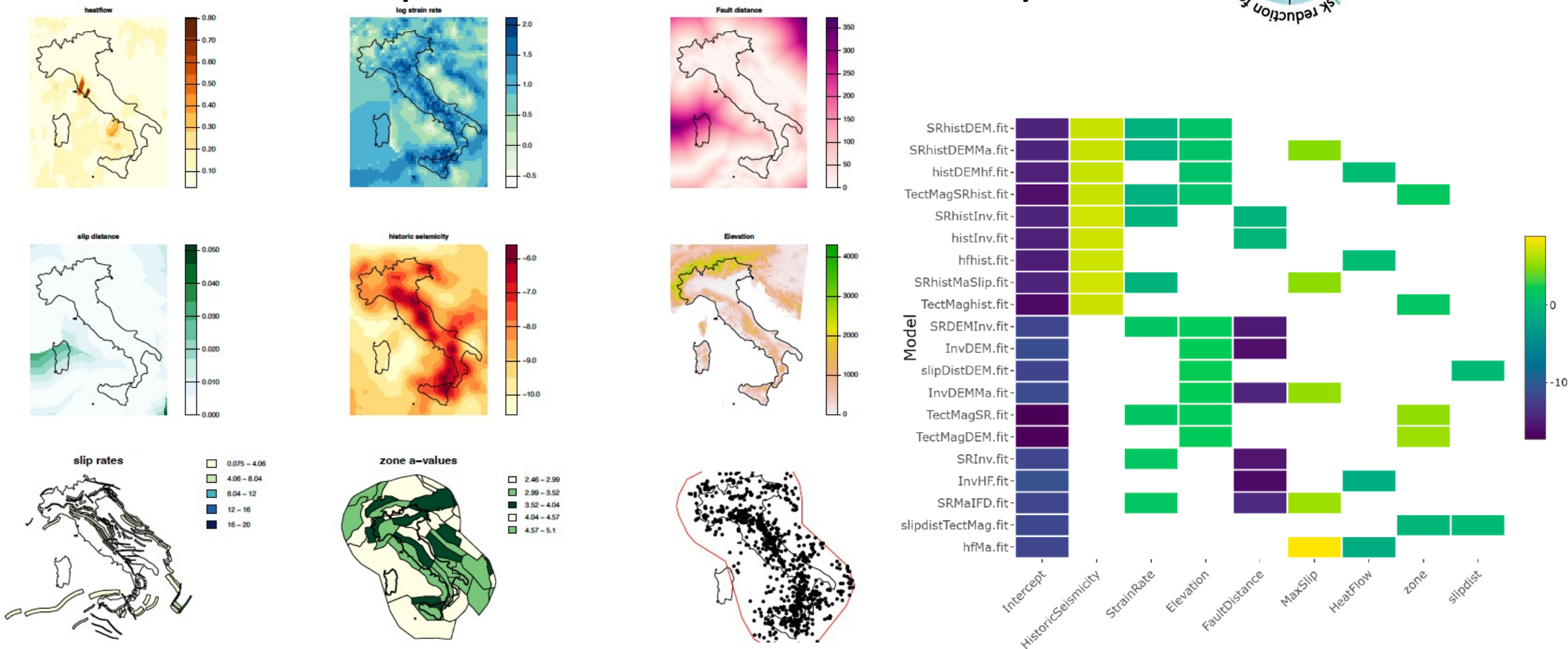
- Full Catalogue
 - NOT declustered

Forecasts of rates for $M > 4.95$

Variance difference shows
how random field is trying
to account for spatial
clustering
(some latent structure)



Time independent – whole Italy



But this is not the best solution because...

... I'm not a fan of declustering.

I'd rather have a background map with an estimate of uncertainty in the clustering process.

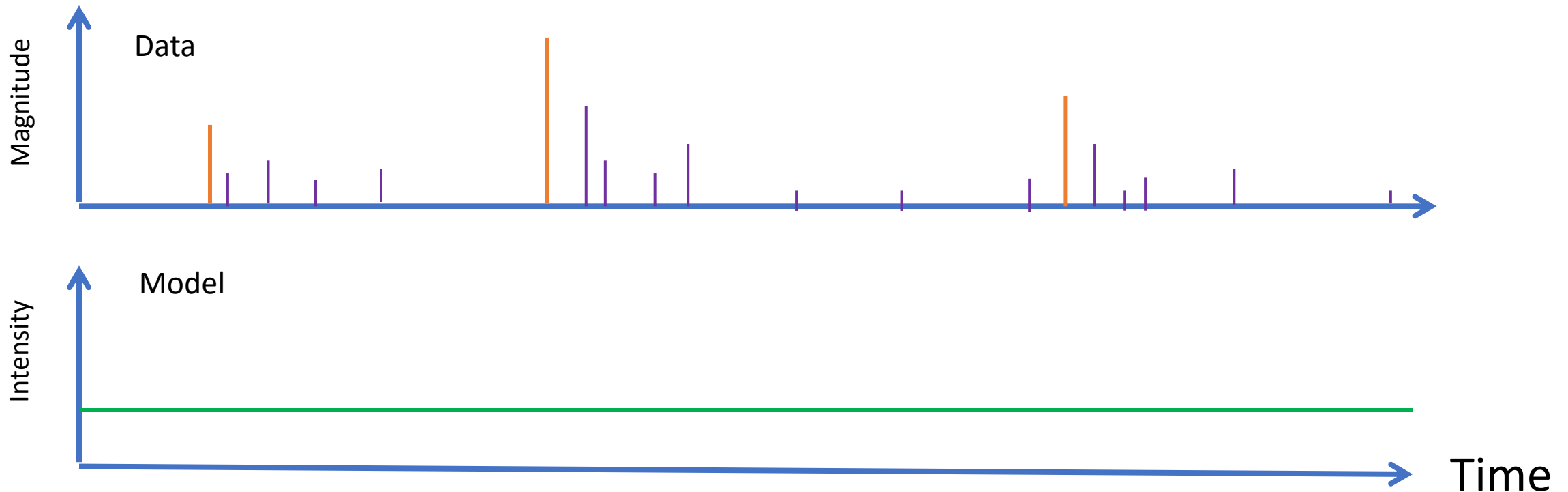
So...

2. Time dependent ETAS like model (Amatrice)



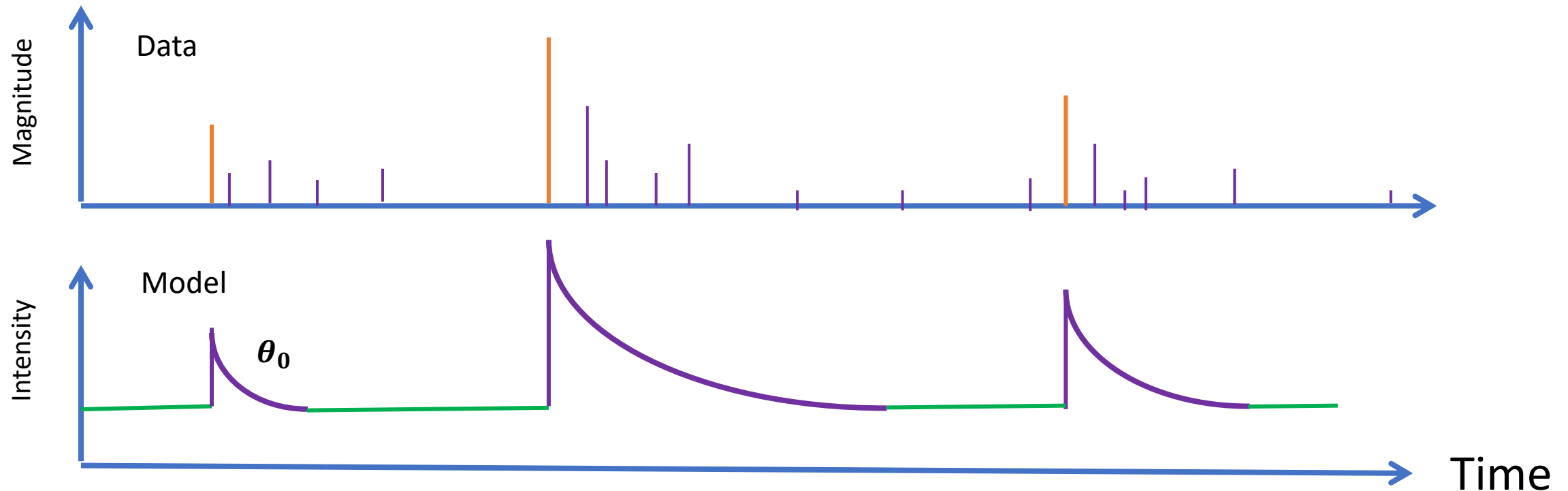
Francesco
Serafini

- Inlabru only had functionality for Log-Gaussian Cox process
- We extended to Hawkes processes (See Francesco's poster)
 - Self exciting point process
 - Trick: lgcp with numerical integration for a sample of ETAS parameters



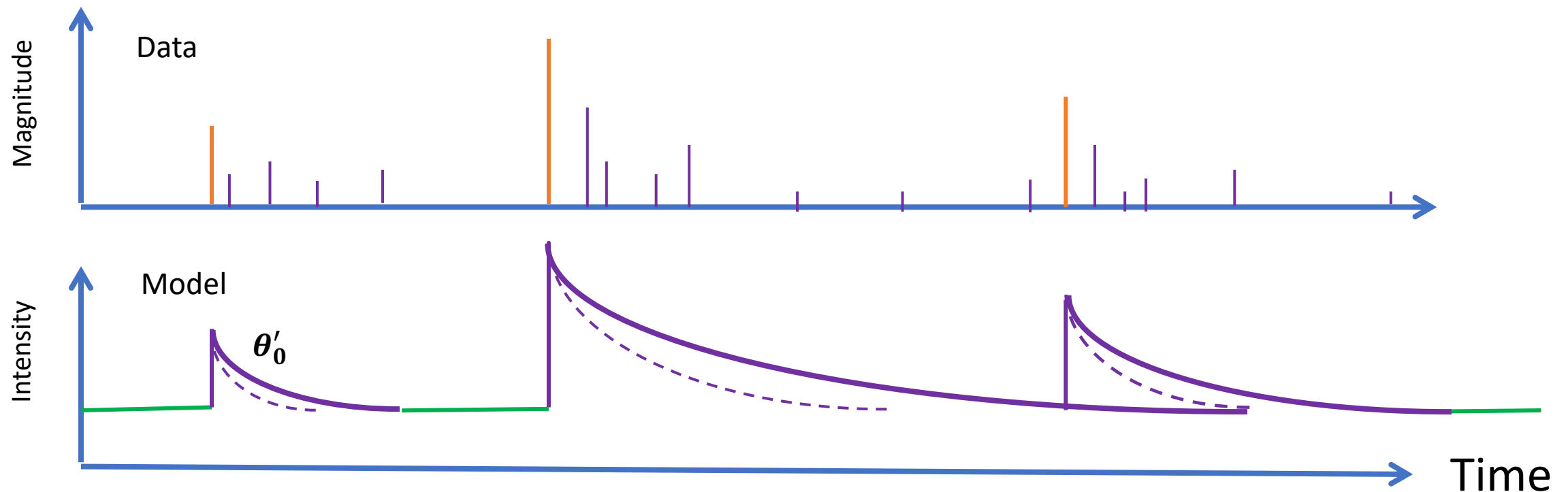
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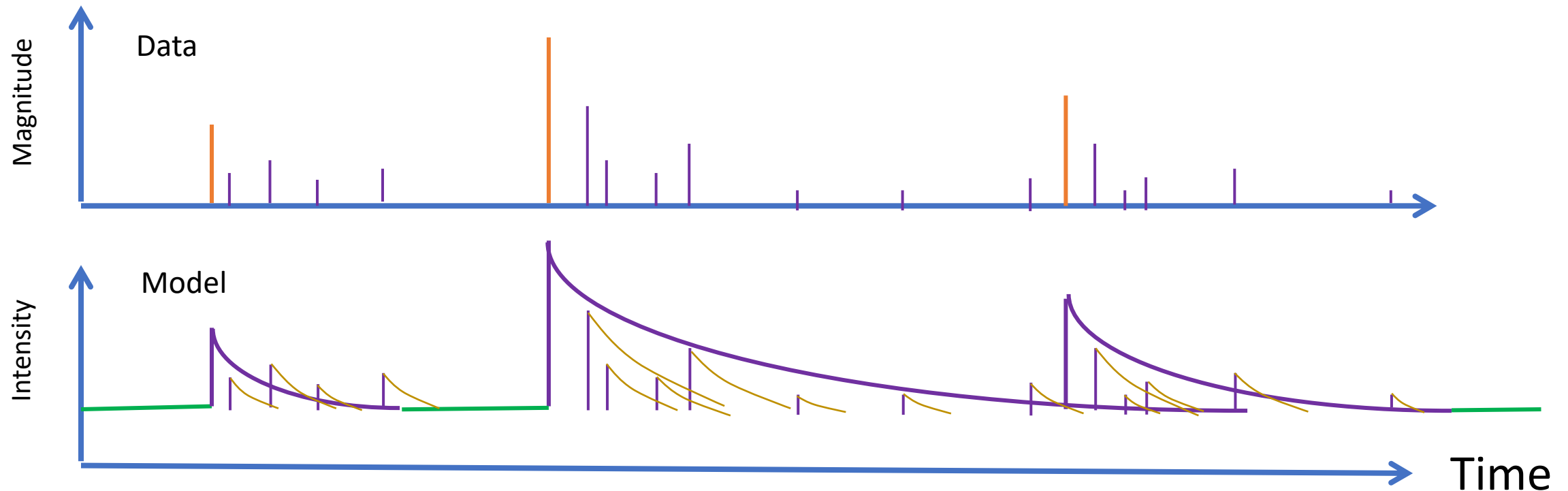
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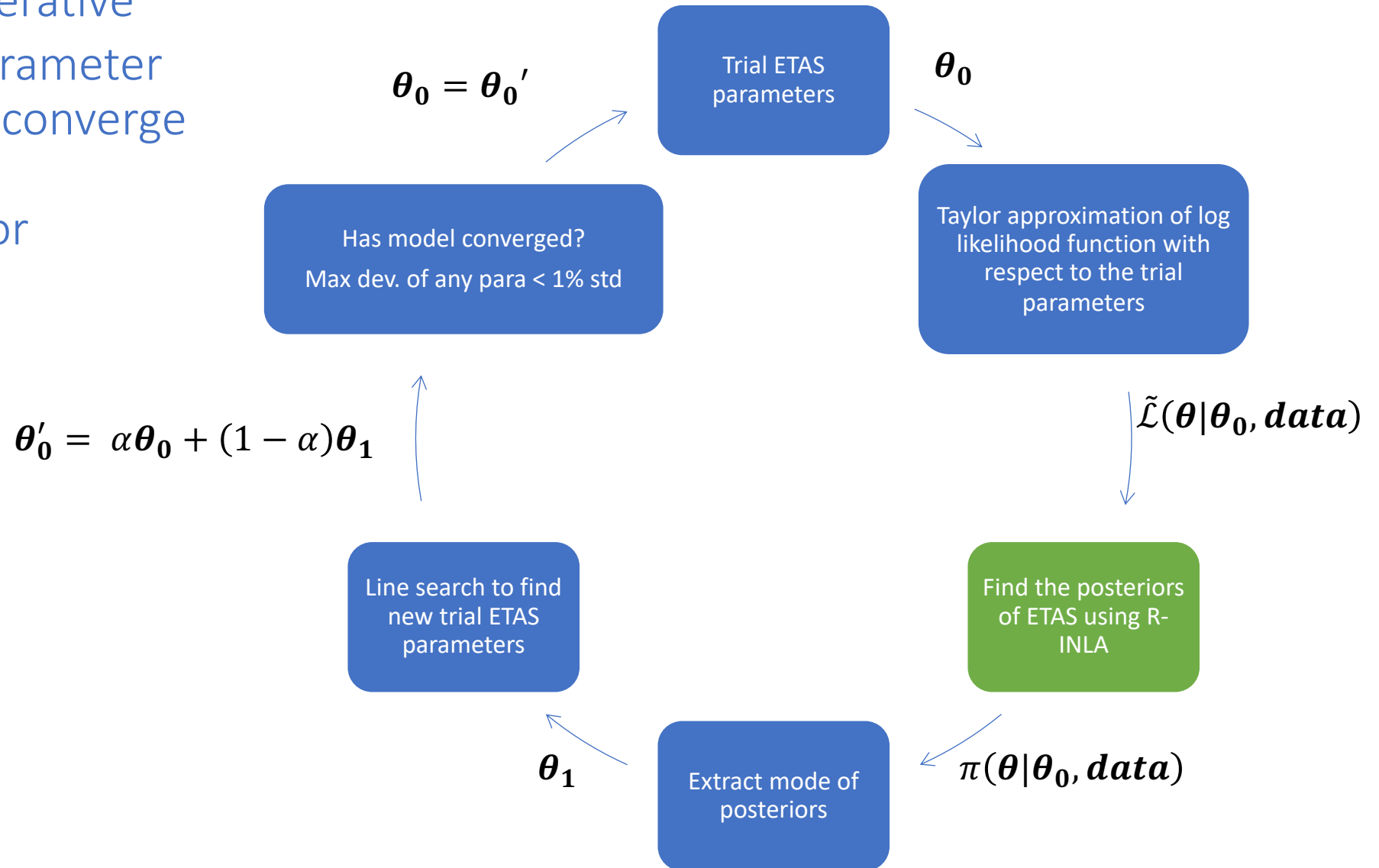
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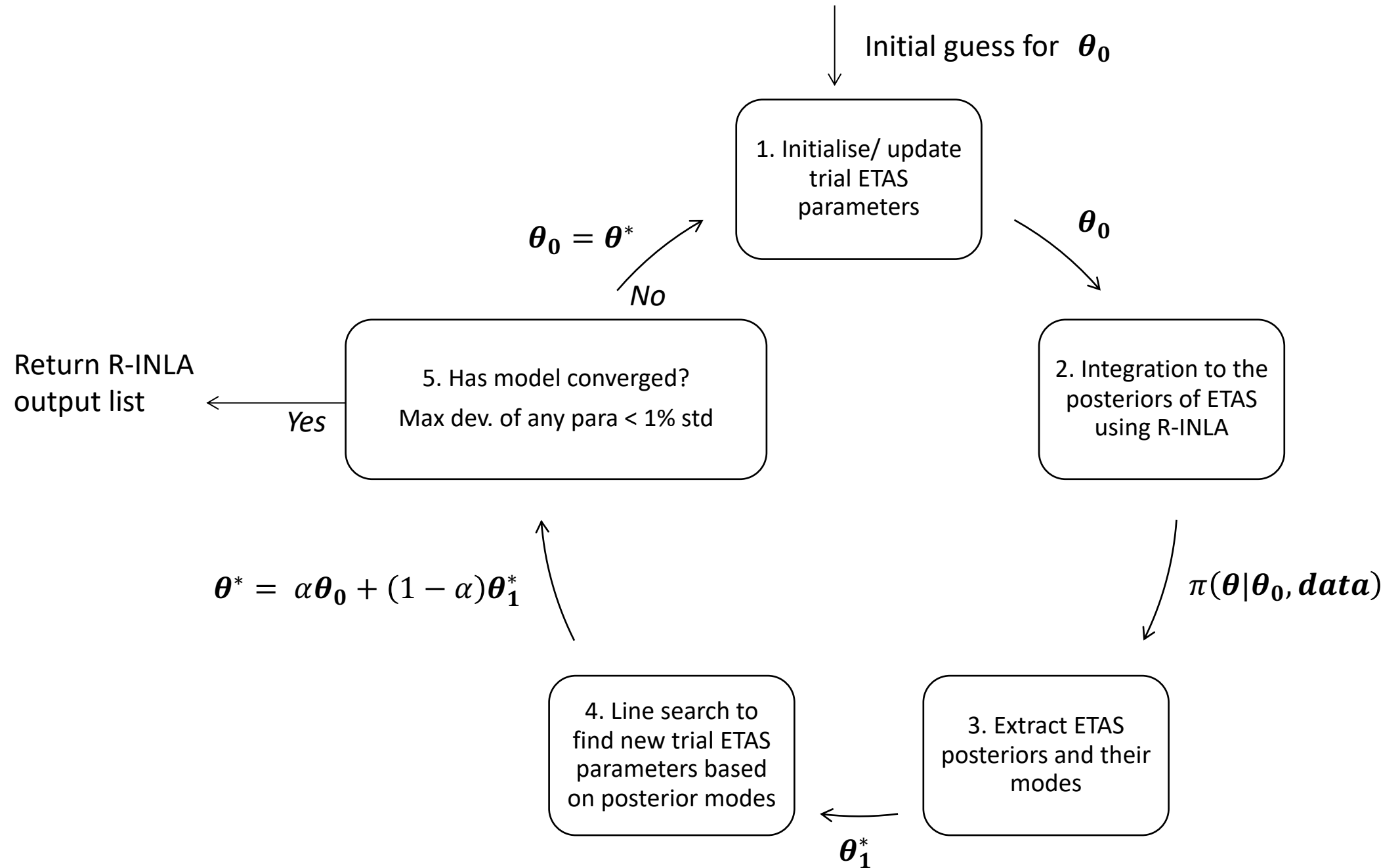


Inlabru

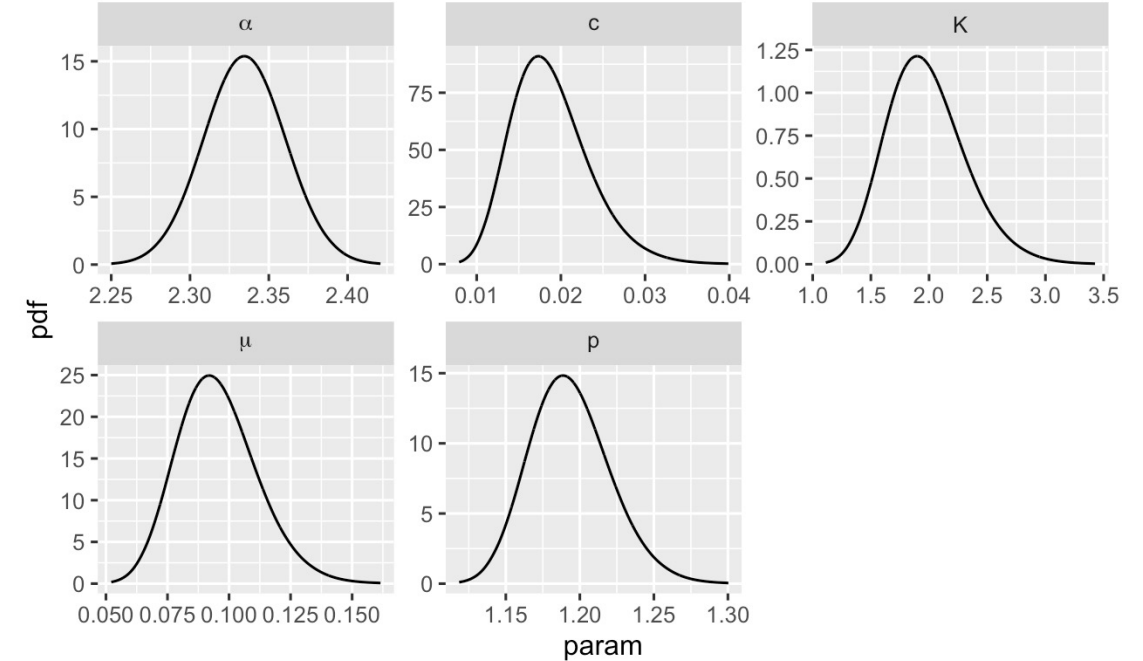
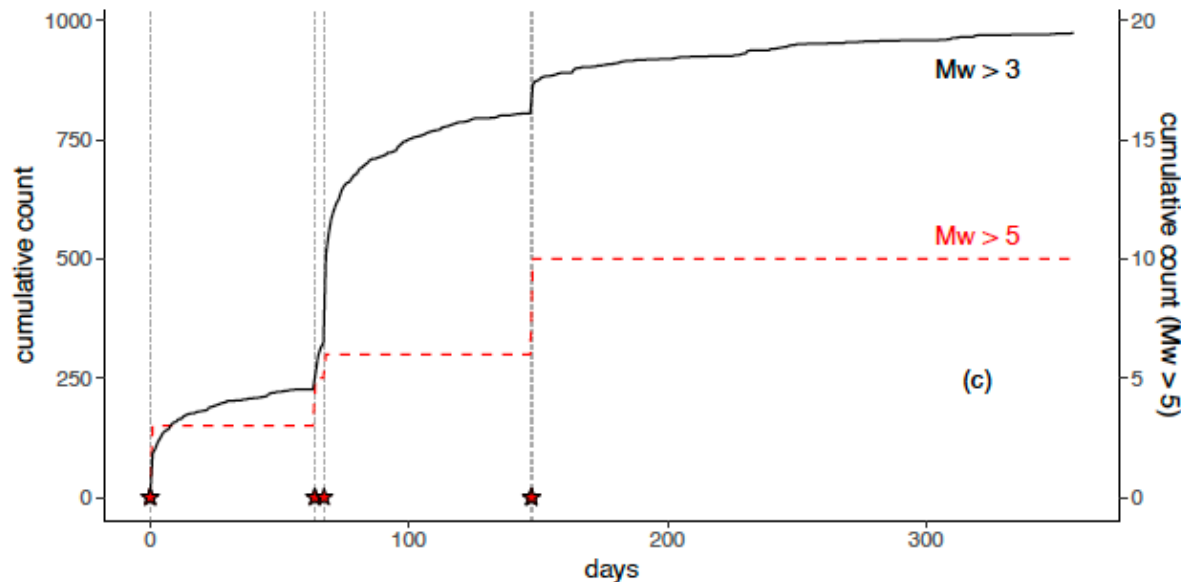
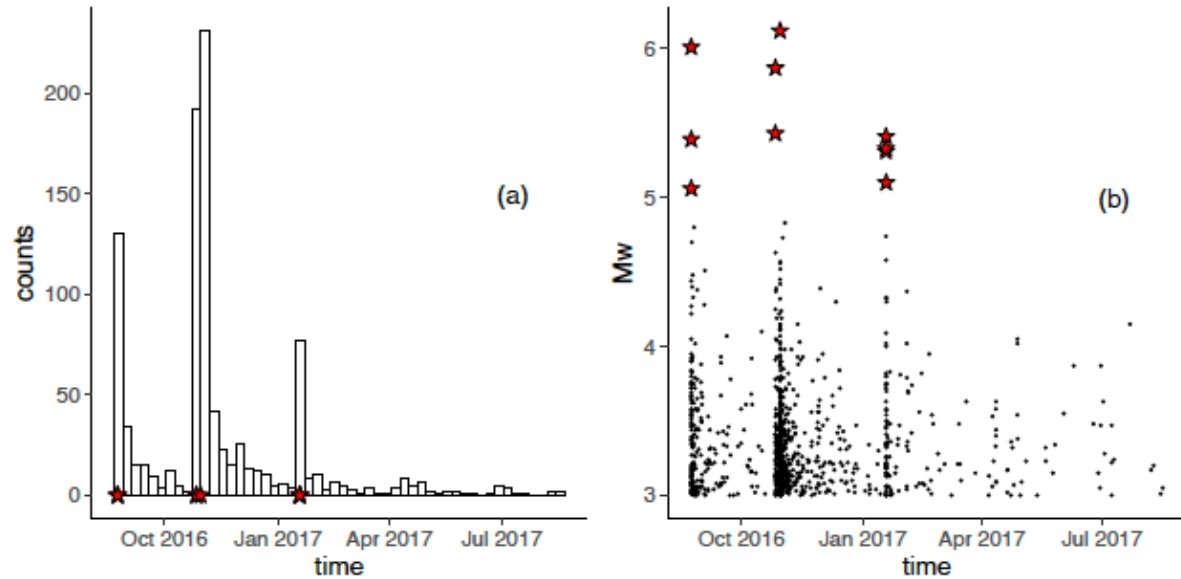
- Manages the iterative update of trial parameter modes until they converge
- Returns posterior distributions



Model initialisation



Sequence: Amatrice

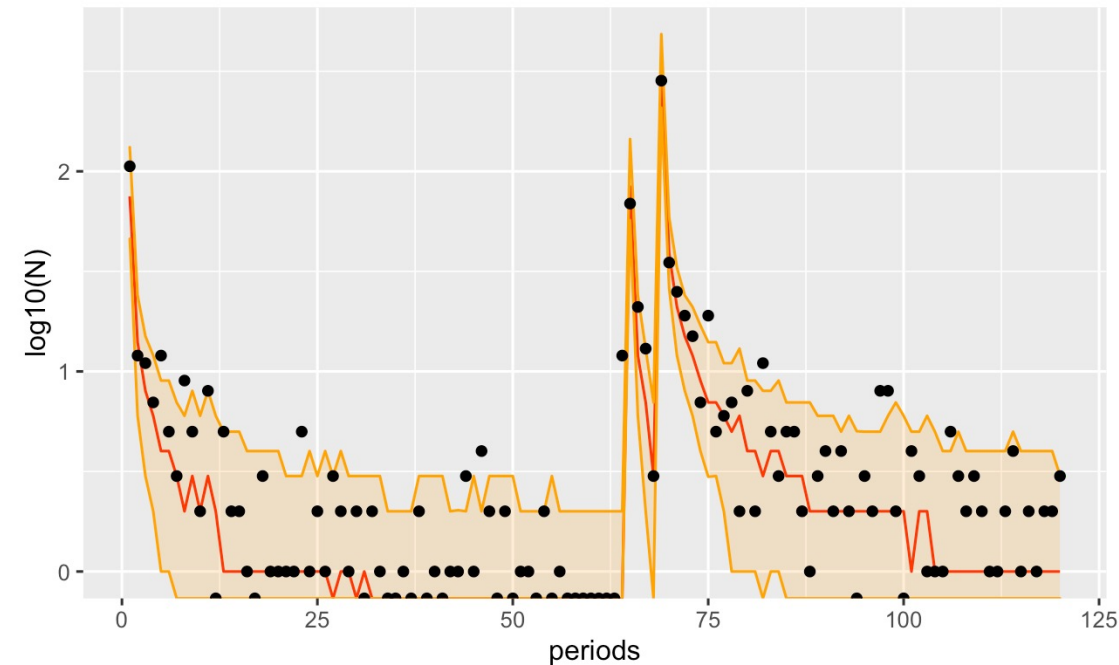
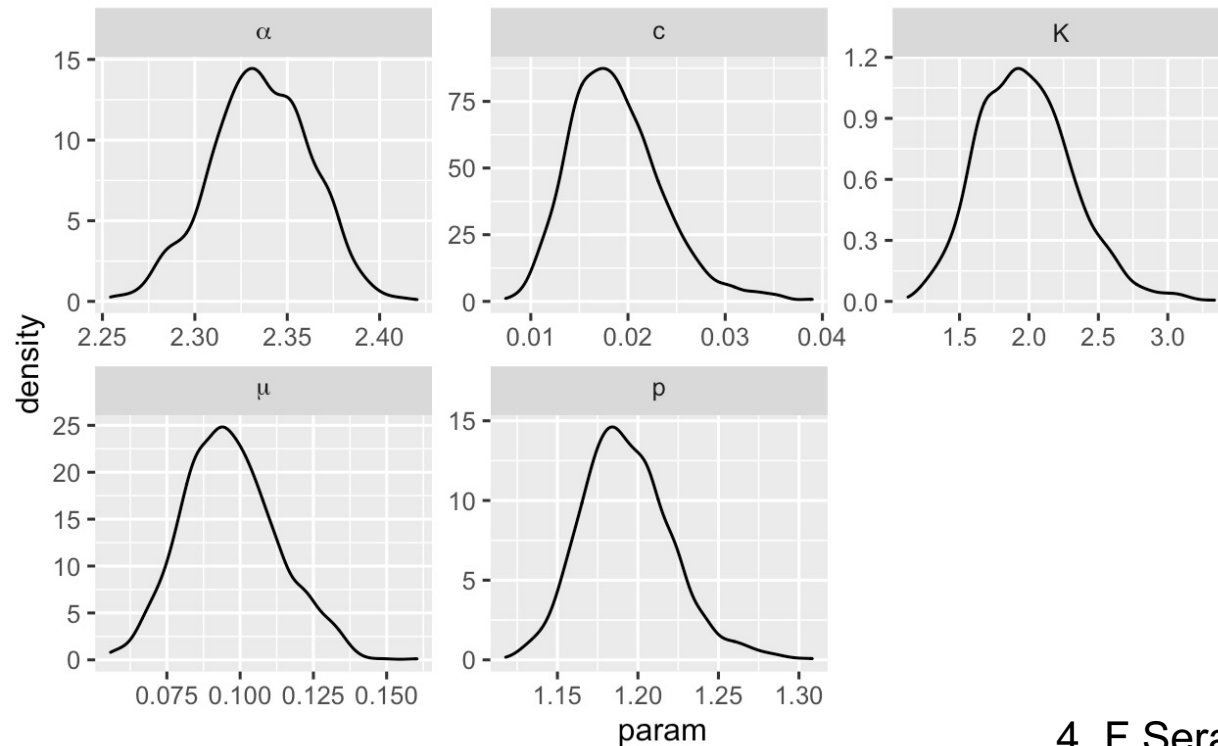


INLA derived posteriors
for ETAS parameters

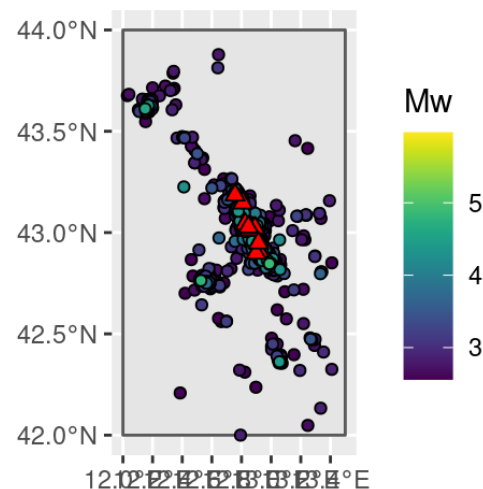
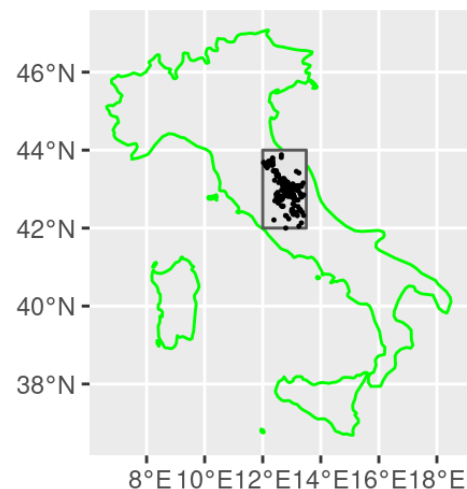
No declustering, jointly optimized

(MCMC fitting often uses a latent variable to partition background and clustered events)

1000 samples from the joint posteriors to generate daily forecasts of following day for $M > 3$ events

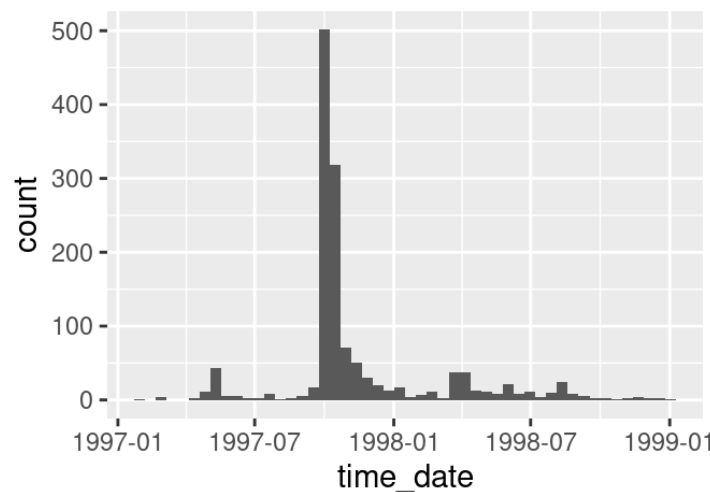
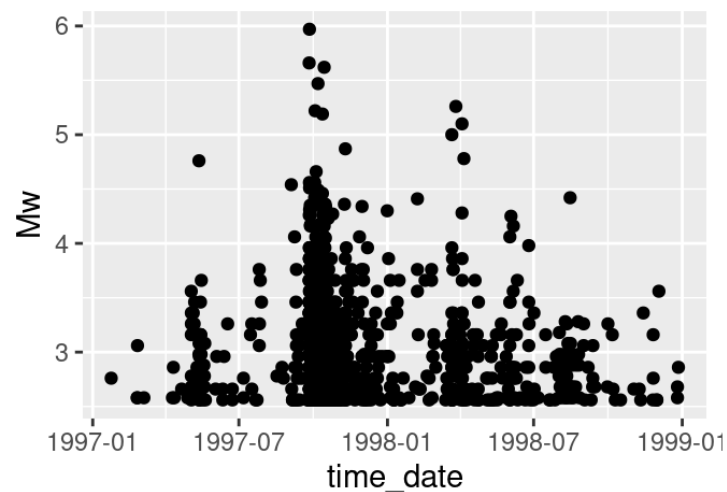


3. Full spatio-temporal modelling: Spatial (*Kirsty*) + Temporal (*Francesco*)

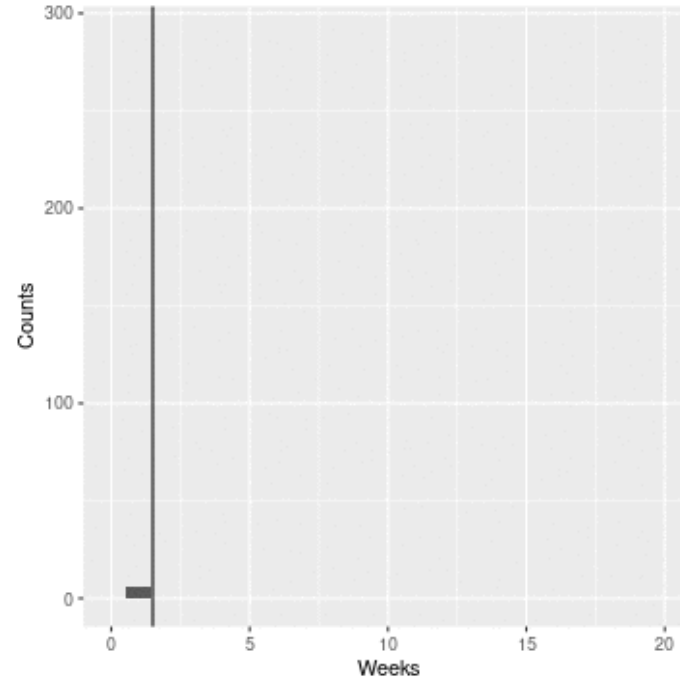
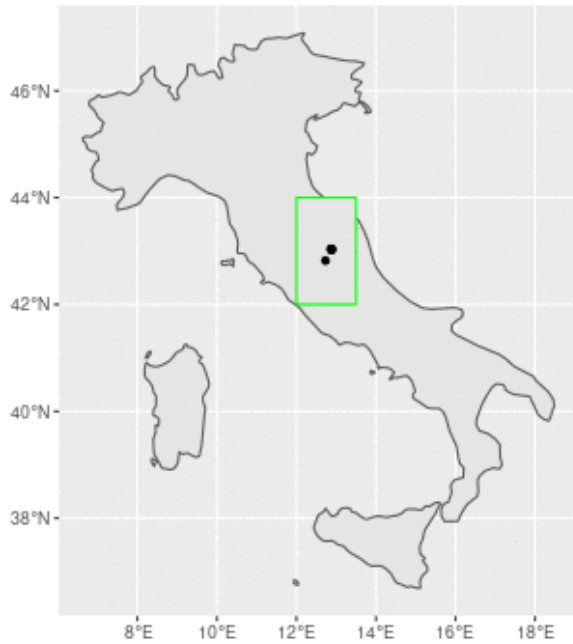


Sequence: Colfiorito

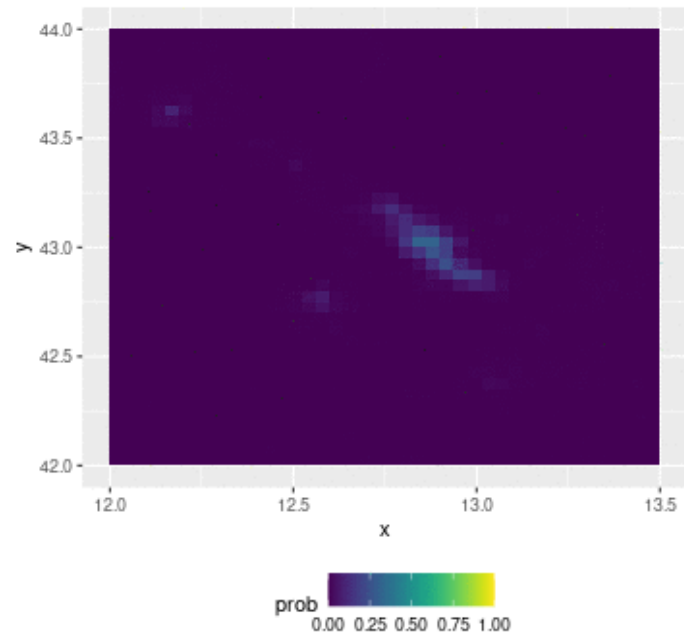
- Horus Catalogue, $M > 2.5$
- 2 years of data
- ~ 1,380 events
- Not (yet) pseudo-prospective



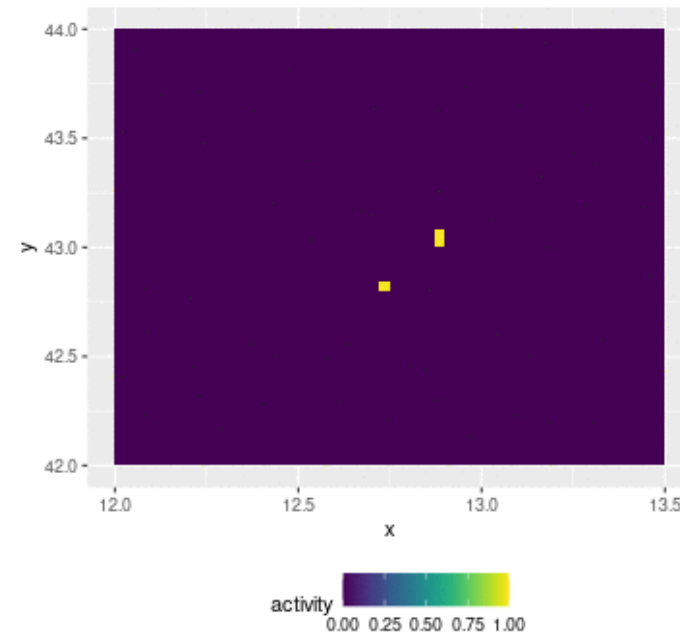
week = 1



Forecast - week 1



Observed - week 1



Weekly retrospective forecast

- Here, the the spatial model is just the random field (work in progress)
- Summary statistics for 10,000 simulations

Some of you might be interested in speed

- Fit retrospective model ~13 minutes on laptop
- 10,000 simulations ~1.5hrs on laptop but...
...it is trivially parallel

Looking forward

- OEF
 - Spatial structure derived from spatial covariates, random field and locally anisotropic kernel (E.g. Ross's talk)
 - Work with available data types and perform hypothesis testing within a consistent framework
 - Forecasts sampling ETAS and spatial covariate posteriors
- PSHA
 - Generate background maps which include uncertainty in the clustering process
 - Expert judgement goes into the priors...
- Exciting opportunities
 - We can let the latent field describe differences rather than drawing boxes around data
 - ETAS parameters and spatial anisotropy linear functions of covariates and random field
 - Surface geology shape files to distinguish evidence of no faults from absence of evidence for faults under basins
 - Explore how quality of forecasts degrades when only some covariates available
 - Censoring for temporal incompleteness
 - Test time dependent covariates (e.g. Coulomb maps)
 - ...

Workflow

2D time-independent seismicity modelling with *inlabru*

Main steps

Main tasks

Sub-tasks

1 Data import

Data Modification

- loading real data *OR*
- generating synthetic data
- converting to `SpatialPoint`
- attaching CRS

```
sp::coordinates(catalogue) <- c("longitude", "latitude")
proj4string(catalogue) <- CRS(SRS_string = "EPSG: ...")
spTransform(catalogue, local_EPSG_string)
fm_crs_set_lengthunit(catalogue, "km")
```

2 Mesh building

Mesh boundary

- based on observed points *OR*
- using a specified polygon

```
inla.nonconvex.hull(coordinates(loc), ...) OR read.delim(...) / read.csv(...) / readOGR(...) / etc.
```

Mesh structure

- tuning mesh quality parameters

```
mesh <- inla.mesh.2d(boundary=..., max.edge=..., min.angle=..., max.n=c..., max.n.strict=..., cutoff=..., crs=...)
```

Mesh assessment

- checking number of vertices
- histogram and map of stdev

```
mesh$n
inla.mesh.assessment(mesh, spatial.range = ..., alpha = ..., dims = ...)
```

3 Specifying model

SPDE model for RF

- setting priors of range and stdev

Linear predictor for RF

- SPDE + intercept

```
pcmatern <- inla.spde2.pcmatern(mesh, prior.sigma = c(...), prior.range = c(...))
components <- coordinates ~ Smooth(coordinates, pcmatern) + Intercept(1)
```

Linear predictor for spatial covariates

- factor covariates
- continuous covariates
- distance sampling data

4 Fitting LGCP model

Fitting LGCP function

- assessing range, stdev, and β

posterior distributions

- plotting range, variance, Matérn covariance & correlation, etc.

```
fit <- lgcp(components, data, domain, samplers, etc.)
```

```
plot(spde.posterior(fit, "Smooth", what = "..."))
```

5 Predicting intensity

Predicting intensity

- Plotting predicted (log) intensity on a pixel map

```
predict(fit, pixels(mesh, mask=...), ~(Smooth + Intercept))
```

6 Comparing models

Estimating abundance

- posterior mean distribution

```
predict(fit, ipoints(boundary, mesh), ~ sum(weight * exp(Smooth + Intercept)))
predict(fit, ipoints(boundary, mesh), ~ data.frame(N=..., dpois(..., lambda = sum(weight * exp(Smooth + Intercept))))
```

Ranking models

- DIC or WAIC ranking

```
deltaIC(fit1, fit2, fit3, ...)
```

7 Spatial rate model

extending to full time-independent forecasts

- magnitude distribution on grid using GR
- magnitude distribution from simulated catalogues using TGR

```
csep_grid_wrapper(fit, lgcp_model, boundary, dh, mag_min, mag_max, b_est, mesh)
forecast_sampler(loglambda, boundary, mesh, crs=crs_wgs84, num_events, b_val, m_min)
```

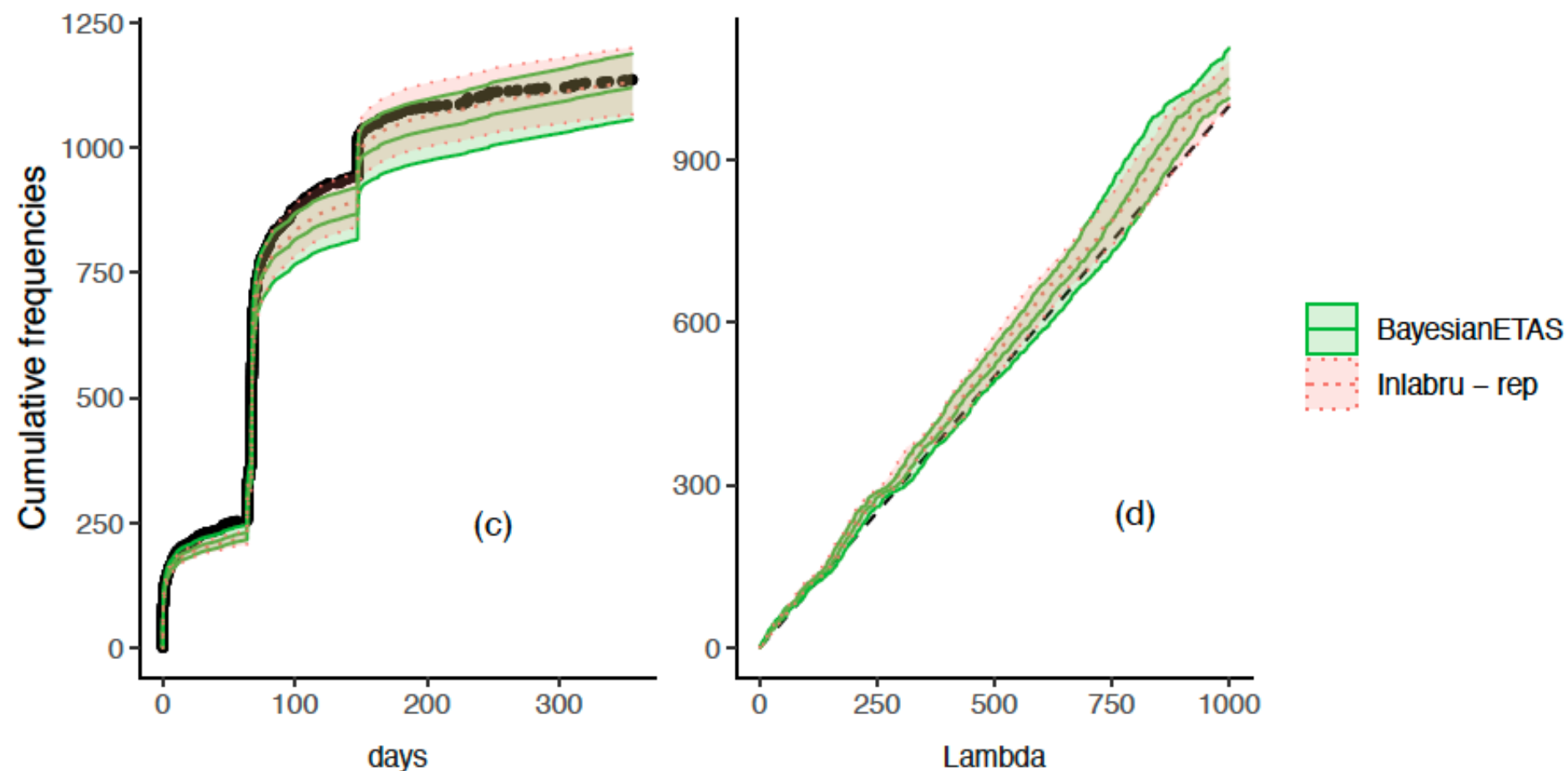
CSEP tests

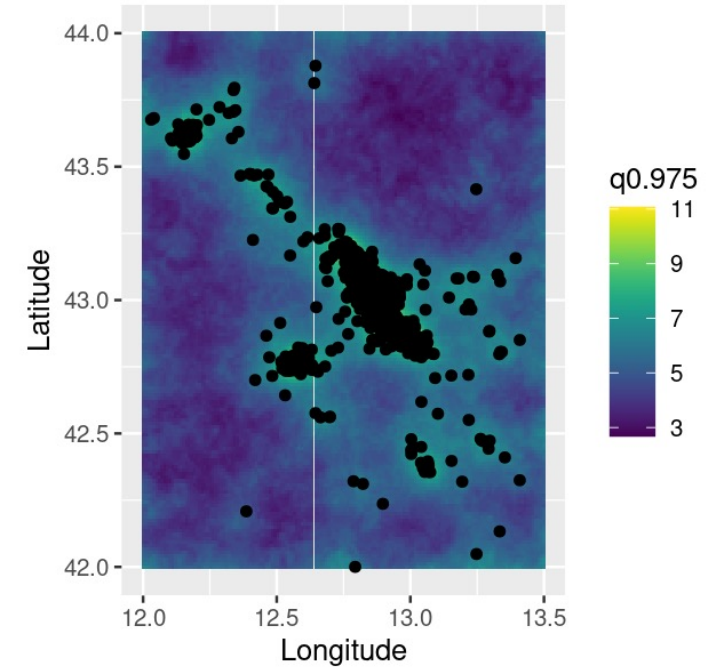
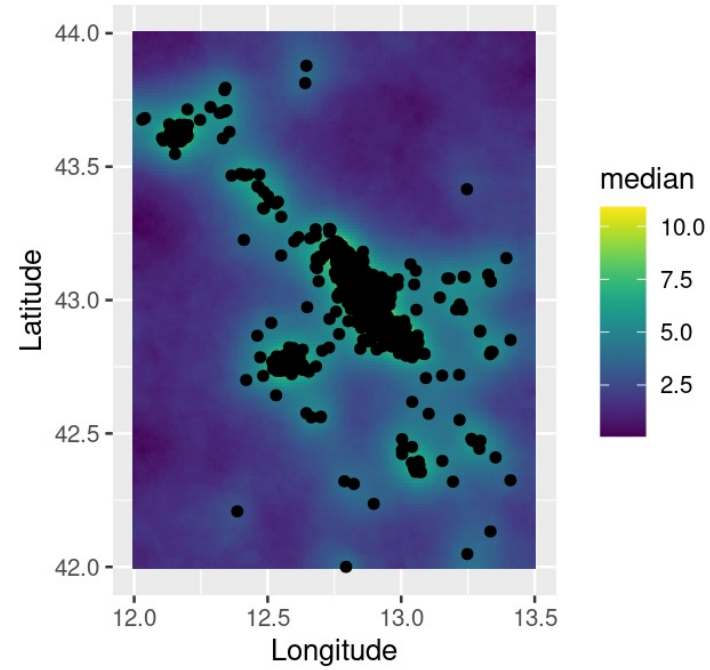
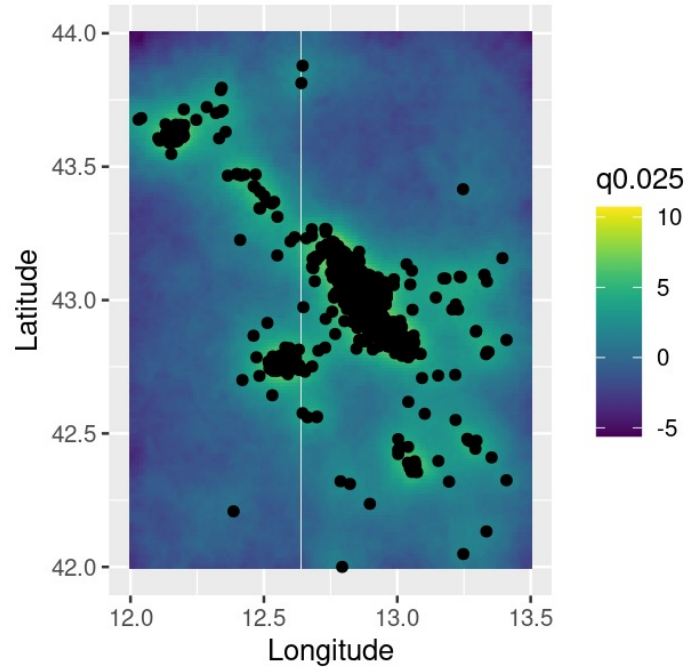
- number test (N-test)
- magnitude test (M-test)
- spatial test (S-test)
- conditional likelihood test (CL-test)
- pseudo-likelihood test (PL-test)

```
pyCSEP package (Savran et al., 2021)
```

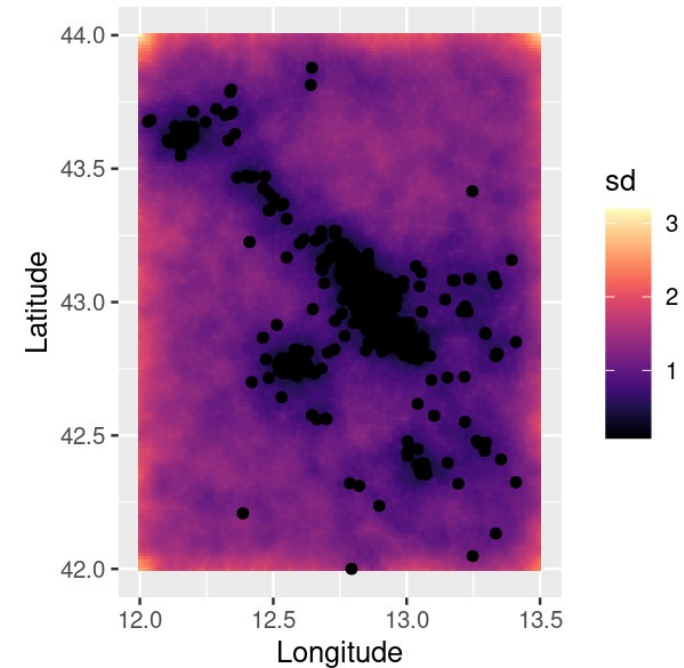
Key functions / codes

Temporal inlabru vs BayesianETAS (Amatrice)

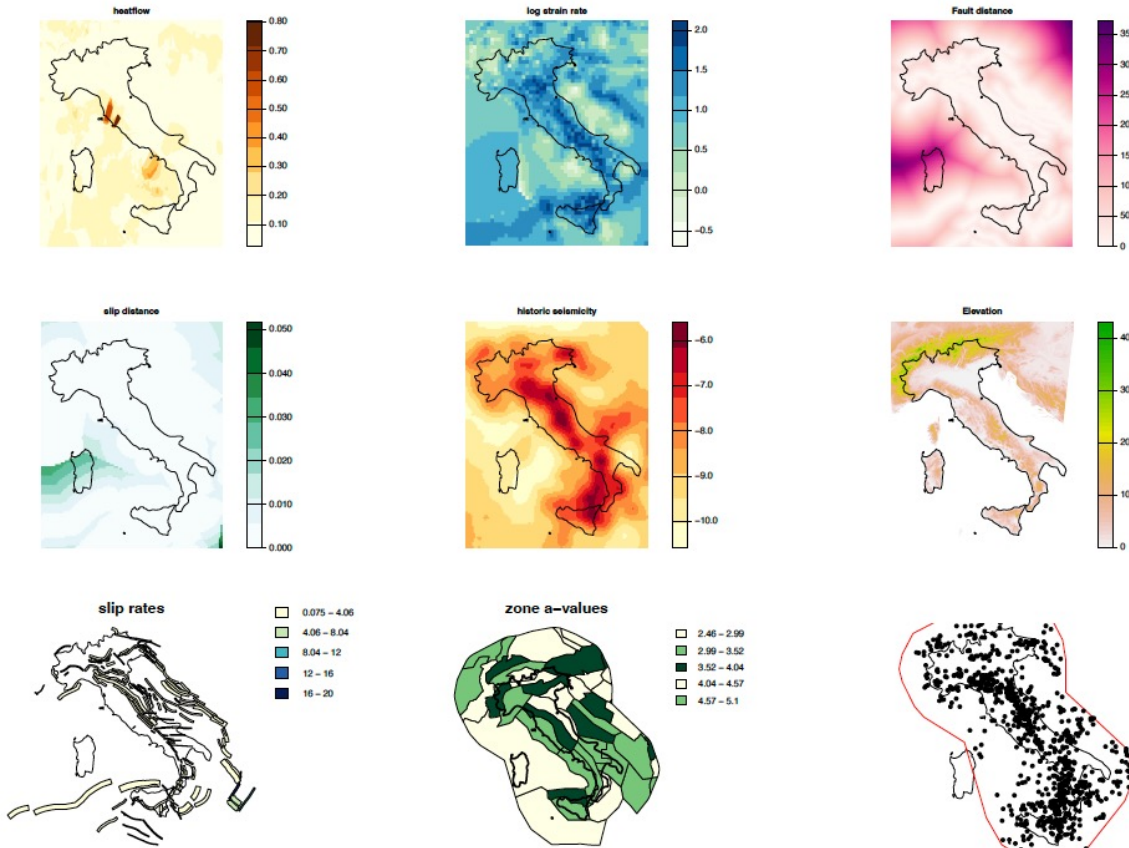




Retrospective Spatio-temporal Colfiorito



Amatrice



Collettini, C., Barchi, M.R., De Paola, N. *et al.* Rock and fault rheology explain differences between on fault and distributed seismicity. *Nat Commun* **13**, 5627 (2022).
<https://doi.org/10.1038/s41467-022-33373-y>

